

GridSense AI

AI-Powered Renewable Generation Forecasting & Decision Intelligence Platform

“From Renewable Energy Forecasts to Intelligent Grid Decisions”

Hackathon: HackOut'26

Theme: Renewable Energy Intelligence

Problem Statement: AI-Powered Renewable Generation Forecasting Platform

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1 Problem Statement

1.1 Renewable Generation Variability

Solar and wind power outputs fluctuate constantly due to weather conditions, time of day, and seasonal transitions. Solar generation is governed by solar irradiance, transient cloud cover, and ambient temperature, while wind generation scales non-linearly with atmospheric pressure fronts and turbulent wind speeds. Unlike conventional controllable thermal generation, renewable power cannot be dispatched on demand.

This physical intermittency creates severe operational difficulty for grid operators and utility companies: *the power grid must commit dispatch schedules and balance supply and demand in advance, relying on imperfect information about weather-dependent generation that will occur 24 to 72 hours later.*

1.2 The Imbalance Challenge: Over-Generation and Under-Generation

Weather fluctuations manifest as two primary operational stress conditions:

- **Under-Generation (Supply Shortfall):** Renewable output falls below scheduled baselines or net demand. Without advance warning, operators must rapidly activate expensive, carbon-intensive fossil peaker plants or resort to emergency load shedding to maintain grid frequency.
- **Over-Generation (Energy Surplus):** Renewable generation exceeds local demand and transmission hosting capacity. If storage or export channels cannot absorb the excess, operators are forced into renewable curtailment, resulting in the permanent loss of clean, zero-marginal-cost electricity.

1.3 Why 24–72 Hour Forecasting Matters

A forward prediction window of 24 to 72 hours provides the critical lead time required for operational coordination:

- **24-Hour Horizon:** Enables optimal day-ahead battery scheduling, spinning reserve procurement, and intraday dispatch adjustments.
- **48–72 Hour Horizon:** Supports medium-term thermal unit commitment, maintenance scheduling, and inter-regional energy trade agreements.

1.4 The HackOut’26 Requirement & Core Problem

The HackOut’26 problem statement mandates a platform that ingests weather data, historical generation records, and site-level parameters to forecast renewable generation over 24–72 hours, identify periods of expected over-generation and under-generation, and recommend suitable grid actions such as curtailment, storage dispatch, and backup activation.

Conventional forecasting tools stop at providing a single, deterministic number (e.g., “*Tomorrow at 14:00: Expected Output = 80 MW*”), leaving operators without uncertainty bounds, risk context, or actionable operational guidance.

Therefore, the core engineering problem is:

“How can renewable generation forecasts be transformed into reliable, uncertainty-aware, and actionable decision support for better grid operations?”

2 Proposed Solution

2.1 What is GridSense AI?

GridSense AI is an AI-powered renewable energy intelligence and decision-support platform that transforms solar and wind generation forecasts into actionable, optimized grid operational decisions.

The platform ingests numerical weather feeds, historical plant generation records, and site-level physical parameters to predict renewable power generation across 24-hour, 48-hour, and 72-hour planning horizons. Rather than terminating at a passive generation chart, GridSense AI quantifies forecast uncertainty, detects upcoming surplus and deficit events, simulates what-if scenarios, and generates explainable dispatch recommendations.

2.2 The Core Intelligence Workflow

GridSense AI bridges the gap between predictive machine learning and control-room operations through a structured eight-stage intelligence pipeline (fig. 1):

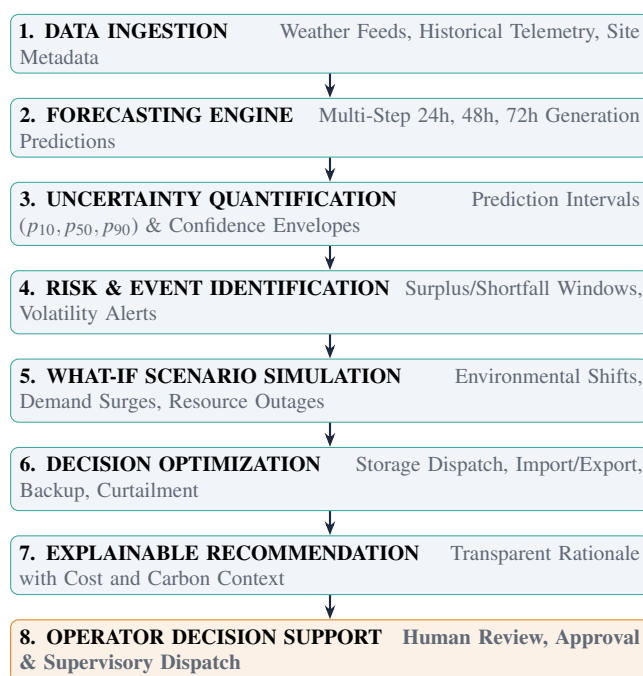


Figure 1: GridSense AI Core Intelligence Workflow: from multi-modal data ingestion to operator decision support.

2.3 Core Distinction: Decision Support vs. Passive Forecasting

The fundamental innovation of GridSense AI is the paradigm shift from passive monitoring to active decision support:

- **Conventional Systems:** Data → Forecast → Passive Dashboard
- **GridSense AI:** Data → Forecast → Risk → Decision → Recommendation

Responsible Decision-Support Positioning: GridSense AI operates strictly as an advisory decision-support system. It assists qualified control-room operators with multi-variable optimizations and transparent justifications, while operators retain final authority over grid dispatch commands.

3 Approach

3.1 Data Collection & Preparation

GridSense AI ingests three primary data feeds specified by the HackOut'26 challenge:

- **Weather Data:** Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), cloud cover percentage, ambient temperature, wind velocity at hub height, and atmospheric pressure.
- **Historical Generation Telemetry:** Timestamped time-series records of past solar and wind output.
- **Site-Level Metadata:** Installed capacity (P_{rated}), geographic coordinates, panel tilt/azimuth, and turbine hub heights.

The ingestion layer performs automated data schema validation, missing-value interpolation for telemetry dropouts, timestamp alignment to standard hourly intervals, and physical boundary verification (e.g., zero irradiance at night).

3.2 Renewable Generation Forecasting Pipeline

For the hackathon prototype, GridSense AI adopts **XGBoost (eXtreme Gradient Boosting)** for multi-step hourly generation forecasting across 24h, 48h, and 72h horizons (fig. 2). XGBoost is selected for its high computational efficiency on tabular data, non-linear interaction modeling, and rapid training cycles.

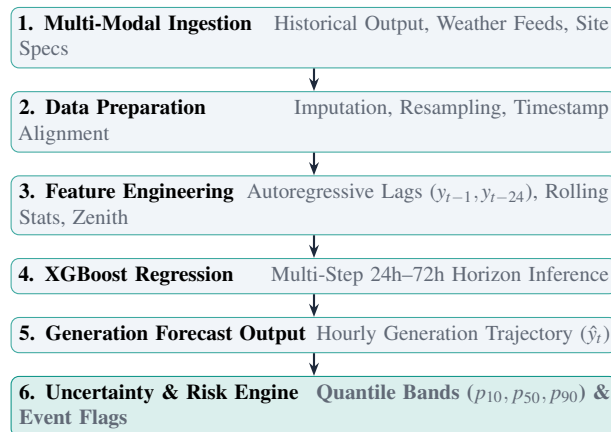


Figure 2: AI/ML Forecasting Pipeline from multi-modal ingestion to uncertainty and risk indicators.

The feature engineering pipeline constructs autoregressive lags ($y_{t-1}, y_{t-24}, y_{t-48}$), 6-hour rolling means and standard deviations, solar zenith angles, and seasonal day-of-year embeddings. Time-aware cross-validation is enforced to eliminate temporal data leakage.

3.3 Uncertainty & Risk Analysis

To prevent misleading single-point assumptions, the platform generates **quantile prediction intervals** (p_{10}, p_{50}, p_{90}). For example:

Expected Generation: 80 MW | Prediction Range: 72–88 MW | Confidence: Moderate

(Note: Numerical values are illustrative examples.)

The system cross-references prediction intervals against scheduled demand to quantify two critical operational risk states:

- **Surplus Risk:** Generation exceeds local hosting and export capacity, creating curtailment risk.
- **Shortfall Risk:** Generation falls below demand baselines, threatening grid stability.

3.4 Scenario Simulation (What-If Sandbox)

GridSense AI provides an interactive what-if simulation sandbox allowing operators to evaluate contingency stress cases:

- **Weather Shifts:** What if cloud cover increases by 30% or wind speeds drop unexpectedly?
- **Demand Perturbations:** What if peak electrical demand exceeds base schedules by 15 MW?
- **Asset State Limitations:** What if battery State of Charge (SOC) is 40% instead of 80%, or a backup unit trips offline?

The system dynamically recalculates forecasts, risk boundaries, and candidate dispatch options in real time.

3.5 Decision Optimization & Action Recommendation

The decision engine evaluates candidate dispatch responses under physical system constraints (fig. 3):

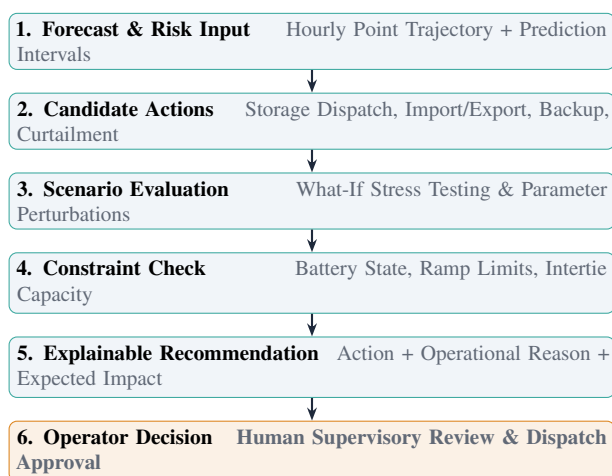


Figure 3: Decision & Optimization operational flow ending in human operator approval.

Candidate actions (BESS dispatch, grid transfers, backup activation, curtailment) are ranked across operational feasibility, cost, and carbon impact. Recommendations follow a clear explainability format:

RECOMMENDED ACTION: Dispatch BESS at 18 MW from 17:30 to 19:30.
OPERATIONAL REASON: Solar generation falls 18 MW below scheduled demand; battery SOC (78%) is sufficient to cover the shortfall.
EXPECTED IMPACT: Eliminates thermal peaker activation, reducing fuel costs and carbon emissions while maintaining grid frequency stability.

The platform acts as an advisory decision-support system; it does not autonomously command physical grid hardware.

4 Key Features

GridSense AI consolidates seven high-value operational capabilities into an integrated control-room platform:

1. **Multi-Horizon Renewable Forecasting (24–72 Hours):** Predicts solar irradiance and wind power generation across 24h, 48h, and 72h horizons by ingesting numerical weather predictions, historical generation records, and site physical parameters.
2. **Probabilistic Uncertainty Quantification:** Generates statistical prediction intervals (p_{10}, p_{50}, p_{90}) and confidence ribbons alongside point estimates, enabling operators to differentiate between stable atmospheric regimes and high-volatility transitions.
3. **Automated Surplus and Shortfall Detection:** Continuously compares generation trajectories against scheduled demand baselines to detect upcoming supply deficits (shortfalls) and excess energy periods (surpluses) with advance warning.
4. **Interactive What-If Scenario Sandbox:** Empowers control-room operators to perturb environmental conditions (e.g., +30% cloud cover), demand spikes, or asset availability (e.g., lower battery SOC or generator outages) and immediately inspect updated forecasts and dispatch options.
5. **Intelligent Action Recommendations:** Synthesizes candidate dispatch strategies—including Battery Energy Storage System (BESS) charging/discharging, inter-regional grid power transfers, backup unit commitment, and managed curtailment.
6. **Explainable AI Decision Support:** Delivers natural-language justifications for every recommended action, explaining the meteorological drivers, risk factors, and expected operational impacts to eliminate black-box hesitation.
7. **Cost and Carbon Optimization Intelligence:** Evaluates operational choices against marginal fuel costs of thermal backup, battery cycling degradation, and avoided carbon emissions, prioritizing zero-emission, low-cost mitigations.

These capabilities operate cohesively under the guiding operational principle:

FORECAST → UNDERSTAND → SIMULATE → OPTIMIZE → ACT

5 Technology & Implementation Plan

5.1 Technology Stack

The GridSense AI prototype is built on a modern, high-performance, and extensible technology stack:

- **Frontend UI: React + Tailwind CSS** for a responsive, modular control-room interface.
- **Visualization: Recharts / D3.js** for high-performance interactive time-series curves and uncertainty ribbons.
- **Backend REST API: Python + FastAPI** providing asynchronous endpoints for telemetry ingestion, forecast generation, and scenario evaluation.
- **AI/ML Engine: Python + XGBoost** executing multi-step tabular time-series regression and quantile inference.
- **Decision Engine:** Python-based rule and multi-objective optimization logic.
- **Data Store: PostgreSQL / JSON Cache** managing historical time series, weather feeds, and scenario state.

5.2 System Architecture & Components

Figure 4 illustrates the modular multi-tier system architecture:

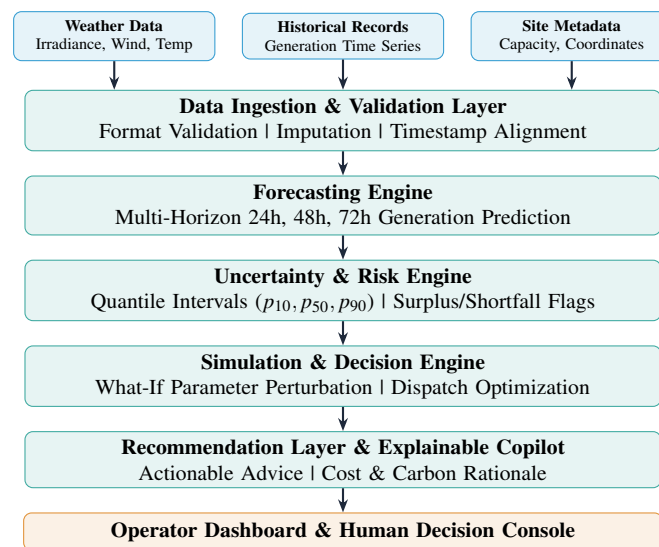


Figure 4: GridSense AI System Architecture: layered progression from data ingestion to operator dashboard.

5.3 Implementation Flow & Prototype Scope

End-to-End Implementation Flow:

Input Data → Preparation → Feature Pipeline → Forecasting → Uncertainty & Risk → Scenario Sandbox → Decision Optimization → Explainable Recommendation → **Dashboard**

The prototype implementation executes across three tightly integrated execution stages:

1. **Data Pipeline & Forecasting Service:** Upon receiving raw telemetry and NWP weather feeds, the ingestion service performs timestamp synchronization and linear interpolation for missing values.

The feature engineering module constructs time-lag and rolling window metrics before dispatching the feature vector to the trained XGBoost model, which returns expected generation curves along with p_{10} and p_{90} uncertainty bounds for the 24–72 hour horizon.

2. **Decision & What-If Execution Core:** The decision service scans the forecast trajectory against anticipated baseline demand to detect potential deficit or curtailment events. If an operator adjusts a what-if slider (e.g., simulating a sudden 30% irradiance drop or battery outage), the engine recalculates risk metrics and ranks dispatch actions based on cost, carbon intensity, and feasibility.
3. **Explainable Dashboard Integration:** The FastAPI backend serializes time-series arrays and structured recommendation payloads to the React interface, rendering dynamic uncertainty ribbons, alert banners, and explainable decision cards for human operator review.

Hackathon Prototype Scope (MVP): The hackathon submission focuses on a self-contained, working end-to-end loop: ingesting real-world solar and wind datasets, predicting generation with confidence intervals across 24–72 hours, detecting imbalance risks, simulating interactive what-if perturbations, and displaying explainable decision recommendations on the operator console.